TrES-5b

R-0677

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_260419` · created 2026-07-27T19:07:14+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461149.9037 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.052
Transit depth (Rp/Rs)^2	3.12 +/- 1.84 [%]
Orbital Inclination (inc)	89.5 +/- 2.88
Airmass coefficient 1 (a1)	1.021 +/- 0.078
Airmass coefficient 2 (a2)	-0.01 +/- 0.038
Scatter in the residuals of the lightcurve fit is	7.65 %
Best Comparison Star	#1 - [258, 429]
Optimal Aperture	1.73
Optimal Annulus	11.45
Transit Duration (day)	0.0871 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.052 vs 0.142 (deviation 0.67σ)
Duration fit vs published	0.0871 d vs 0.0758 d (deviation 1.76σ)
Mid-time O–C	1.6 min
Depth SNR	3.4
Residual scatter	7.65 %

Light curve

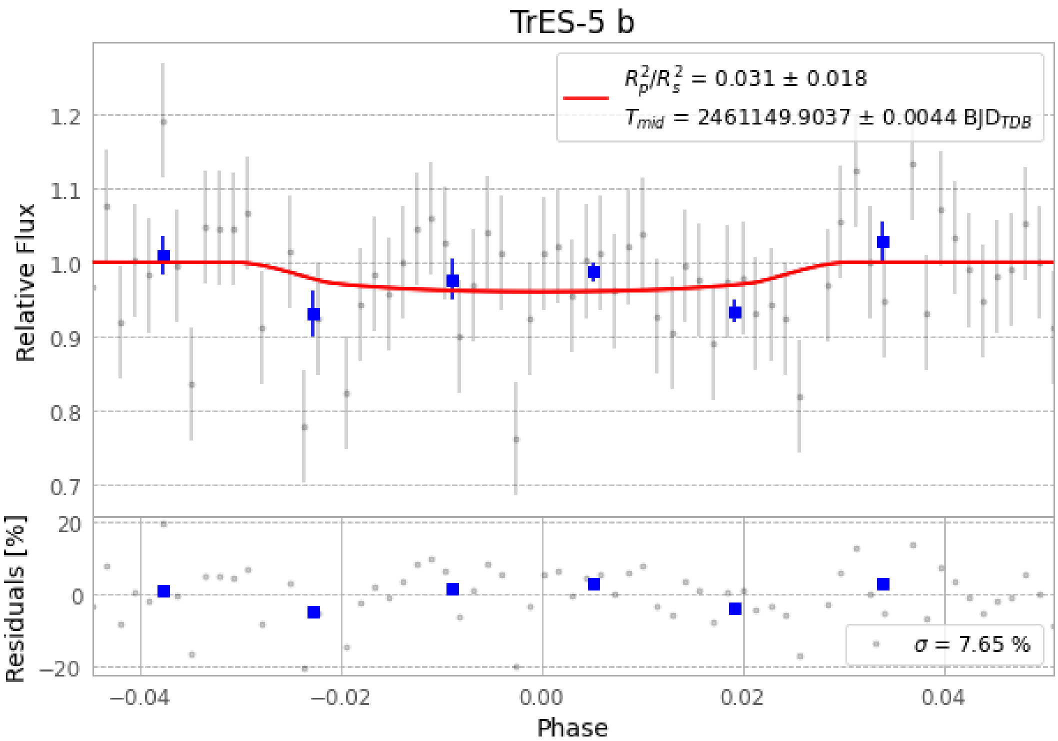

Figure 1 — FinalLightCurve_TrES-5 b_2026-04-19.png (SHA-256 88992a62c1229721…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [275, 291], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 7d038235224acb59…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813`

Data provenance

69 FITS frames (45 MB), `TRES-5260419080616.FITS` → `TRES-5260419113017.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-260419.log (50279909fb7b…), FinalLightCurve_TrES-5 b_2026-04-19.png (88992a62c122…), FinalLightCurve_TrES-5 b_2026-04-19.csv (6a79d23528a7…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

